InsightEngine: An End-to-End Data Engineering Platform for Business Intelligence

Team Members :

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Overview

Abstract

InsightEngine is an end-to-end data engineering platform that collects, ingests, processes, transforms, stores, and analyzes business data from multiple sources. Python is used for data generation, Apache Kafka for streaming ingestion, PostgreSQL for storage, and PySpark for data processing and transformation. Apache Airflow automates and orchestrates the pipeline, while Streamlit provides an interactive dashboard with continuously updated business insights. The project demonstrates the complete data engineering lifecycle, including data ingestion, validation, cleaning, transformation, modeling, and analysis.

Introduction

In today's data-driven environment, organizations generate large volumes of data from sources such as transactions, products, payments, and inventory. This raw data is often distributed and inconsistent, making it difficult to use for decision-making. Data engineering helps collect, store, process, transform, and organize this data into reliable information.

InsightEngine demonstrates the complete data engineering lifecycle by generating business data using Python, supporting batch and streaming ingestion with Apache Kafka, storing data in PostgreSQL, and processing it using PySpark. Apache Airflow automates the pipeline, while dimensional modeling organizes the processed data for analysis. Finally, a Streamlit dashboard presents meaningful and interactive business insights.

Objectives

- **To develop an integrated platform that demonstrates the complete data engineering lifecycle.**

- **To connect the entire data pipeline with a simple interface for consuming business data and insights.**

- **To demonstrate how an integrated data engineering workflow supports data-driven decision-making.**

Literature Survey

Title	Authors	Year	Key Drawback	Future Work	Link
FlowETL: Autonomous Data Engineering Pipeline	Di Profio et al.	2025	Limited generalization	Improve automated ETL	Paper
Scalable ETL with Spark, Airflow & Snowflake	U. Nayak	2025	Configuration complexity	Improve scalability & cost	Paper
Serverless ETL using AWS Glue & PySpark	Vuppu & Achanta	2025	Cloud dependency	Optimize resource usage	Paper
Architecting Resilient ETL Pipelines	J. Nalla	2025	Monitoring complexity	Improve data quality & reliability	Paper
AI-Driven Data Engineering Workflows	Govindarajulunaidu & Narayanan	2025	AI increases complexity	Develop self-optimizing ETL	Paper

Literature Survey

Title	Authors	Year	Key Drawback	Future Work	Link
AI-Powered Data Engineering for ETL	Y. Tandon	2025	Explainability challenges	Intelligent data-quality checks	Paper
AIDEN: AI-Driven ETL Networks	S. R. Bolla	2025	Cloud-focused architecture	Adaptive resource allocation	Paper
Agentic Data Pipelines	N. Mangala	2025	High orchestration complexity	Self-healing pipelines	Paper
ETLT & ELTL Design Patterns	Rucco et al.	2025	Governance complexity	Improve lineage & observability	Paper
Data Engineering with Jayvee DSL	Heltweg et al.	2025	Limited DSL features	Improve pipeline development tools	Paper

Literature Survey

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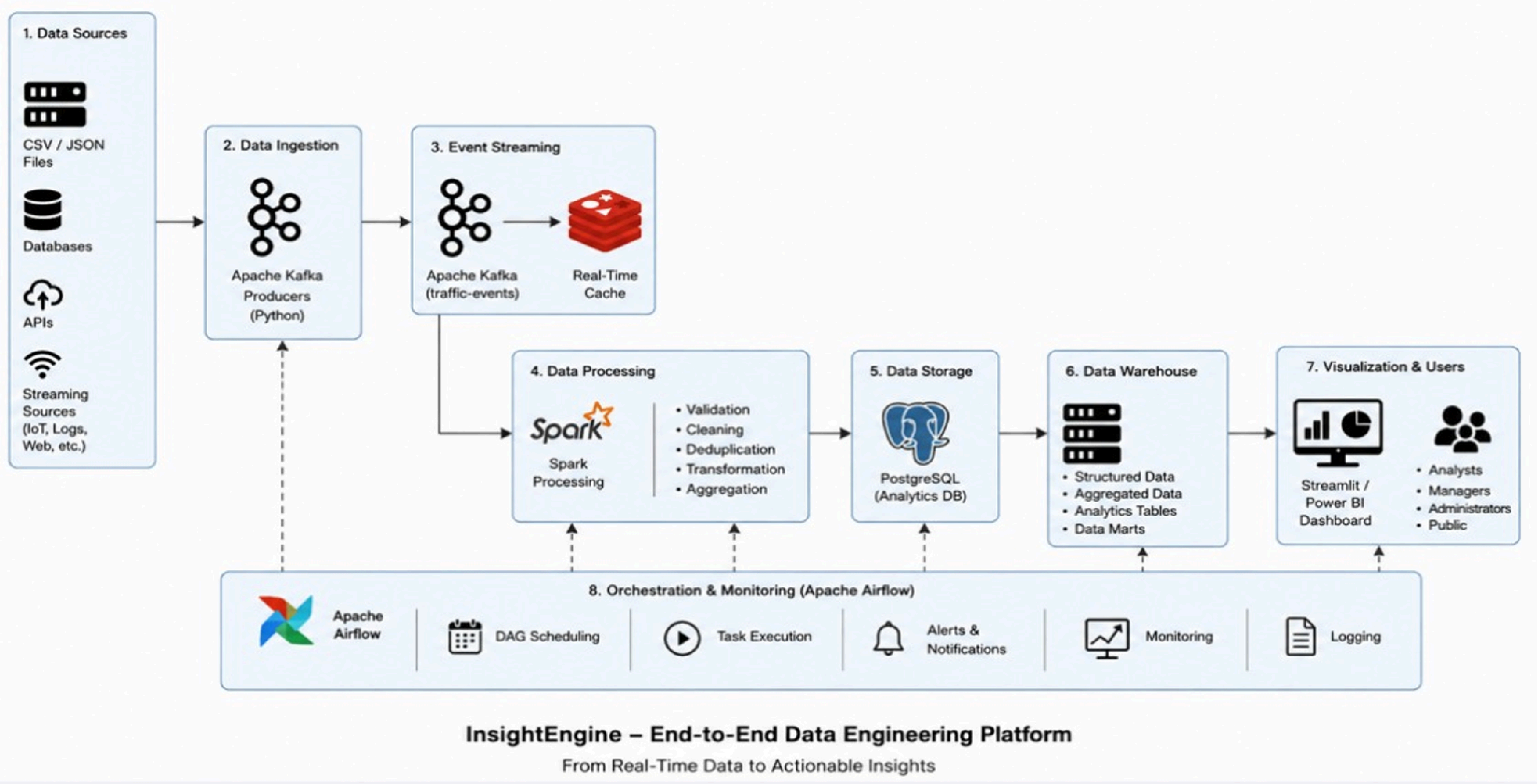
Scalable ETL with Spark & Airflow	U. Nayak (2025)	Configuration complexity	Improve scalability — Link
Serverless ETL with PySpark	Vuppu & Achanta (2025)	Cloud dependency	Optimize resources — Link
Resilient ETL Pipelines	J. Nalla (2025)	Monitoring complexity	Improve reliability — Link
AI-Driven Data Engineering	Govindarajulunaidu & Narayanan (2025)	AI complexity	Self-optimizing ETL — Link

System Architecture



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